Aansh Yadav

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Aspiring computer scientist focused on numerical algorithms, cryptography, and software engineering.

Education

Somerville School, India — Grade 11 (CBSE)

- CBSE Class 10 Board Examinations: **94.8%**

Honors & Awards

- **Lodha Genius Programme '26, Ashoka University** — Selected from 21,000+ applicants (1.4% acceptance) for a fully funded residential research program; trained in applied mathematics, numerical algorithms, cryptography, and number theory.
- **International Psychology Olympiad (IPSYO) 2026** — Highest Distinction, 286/300.
- **Inter-Section Indian Music Competition** — First Place (tabla).
- **Pracheen Kala Kendra** — Sangeet Bhushan 3rd Year, First Division (tabla).

Technical Projects

Newton Fractal Generator — Python, NumPy, SymPy

- Extended the Newton-Raphson method into the complex plane to render Newton fractals.
- Vectorized the iteration across 360,000 starting points per render, coloring each by the root it converges to.

Smart SVD: Matrix Decomposition Studio — Python

- Built an SVD image-compression tool from scratch, designed around human vision.
- Converted images to YCrCb and assigned separate ranks to luma and chroma instead of compressing RGB channels equally.
- Added CLAHE and high-boost filtering to preserve edge sharpness with no added computational cost.

Indian History Quiz Master — Python, Tkinter

- Built a desktop quiz application with a graphical interface.

Voice Verification Security Layer — Web

- Designed a biometric voice-verification layer for my web projects.

Leadership & Activities

Computer Club & Annual Technology Showcase, Somerville School — Core Member; Event Head, Web Development

- Train younger students in web development and help peers move from foundational logic to working implementations.

Tabla, Pracheen Kala Kendra — 6+ years

- Lead Tabla Player for the school's premier classical music showcase.

Competitive Chess

- Tournament player; competed in the 5th MCA Invitational Chess Tournament.

Skills & Certifications

- **Programming:** Python, JavaScript, HTML, CSS
- **Libraries & frameworks:** NumPy, SymPy, PyTorch, Tkinter
- **Coursework & certifications:** Deep Learning with PyTorch: Image Segmentation (Coursera); Leading in the Age of Generative AI (Codecademy)
- **Languages:** German — Goethe-Zertifikat A1 and A2 (70/100), preparing for B1